

```
    return 0;  
}
```

The output of this program is:

1, 2, 3, 4, 5

Dynamic Memory

Dynamic memory is the memory to be used in a program during its run time. C++ provides two effective operators — `new` and `delete`, which are used for this purpose.

The `new` And `new[ ]` Operators

The `new` operator is used for dynamic memory allocation. This operator should be followed by a data type. If more than one element is required, specify it within brackets `[ ]`. The general form of using this operator is as follows:

```
p1 = new type  
p1 = new type [ number_of_elements]
```

Here, the first expression `p1 = new type` is used for allocating memory so that one single element of type `type` can be contained. Similarly, the second expression is used for assigning an array of elements of type `type`. Here, `number_of_elements` is an integer value. Consider the statements:

```
int * b;  
b = new int [ 6];
```

In the second statement, the `new` operator dynamically assigns space for six elements of type `integer` and returns a pointer to the first element of the sequence assigned to `b`. Therefore, the pointer `b` points to a valid block of memory with space for six elements of type `integer`.